

Part I

SUMMARY of Antiderivatives And Area:

Summary

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Summary

We know, from the Antiderivatives section, that

(1) the position function, $s(t)$, of the object moving along a straight line with velocity $v(t)$ is an antiderivative of the velocity, i.e.

$$\int v(t) dt = s(t) + C$$

(2) the displacement of the object over the time interval $[a, b]$ is given by

$$\text{Displacement} = \Delta s = s(b) - s(a)$$

In this section, we learned that

(3) the displacement of the object over the time interval $[a, b]$ can be represented as the area under the curve $y = v(t)$, (if $v(t) \geq 0$ on $[a, b]$), which means that

$$\text{Displacement} = \int_a^b v(t) dt$$

(4) Combining the last two equations, we get

$$s(b) - s(a) = \int_a^b v(t) dt$$

Note: The last formula states that a definite integral of a *velocity* function can be computed by using an antiderivative of the function and evaluating it at the end points of the interval. This is the first time that we make a connection between definite integrals and antiderivatives.

Part II

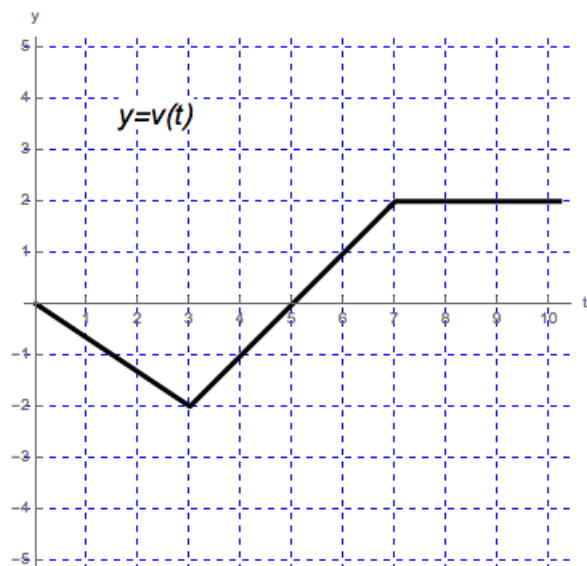
Recitation Questions

Problem 1

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Problem 1

Problem 1 The graph of the velocity (t in min, v in ft/min) of a particle moving along a straight line is given in the figure. Assume that the particle was at the origin initially.



- (a) Evaluate the displacement of the particle over the following intervals.
- (i) $[0, 10]$
 - (ii) $[0, 7]$
 - (iii) $[0, 5]$
- (b) When was the particle farthest from the origin? [3]
- (c) What was the total distance the particle has travelled over the time interval $[0, 10]$?

Problem 2

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Problem 2

Problem 2 The velocity of an object moving along a straight line is given by $v(t)$ (in ft/s) and we only know the following: $\int_0^3 v(t) dt = -3$, and $\int_3^4 v(t) dt = 5$.

Compute the following displacements, if possible. If it is not possible, give examples explaining why not.

- (a) $s(3) - s(0)$
- (b) $s(4) - s(0)$
- (c) Find the displacement during the interval $[0, 4]$ if the velocity at the time t , $0 \leq t \leq 4$, was $v(t) + 2$ ft/s instead?
- (d) Find the displacement during the interval $[0, 4]$ if the velocity at the time t , $0 \leq t \leq 4$, was $5v(t)$ ft/s instead?

Problem 3

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Problem 3

Problem 3 The velocity of the object moving along a straight line is given by $v(t) = t - 5$, $0 \leq t \leq 10$, where t is in seconds and v in ft/s.

- (a) Use geometry to evaluate the displacement of the object on the time interval $[0, 10]$. Sketch the graph of the velocity and shade the relevant region.
[3]
- (b) Compute the displacement on the interval $[0, 10]$ using antiderivatives of the function v .
- (c) Use geometry to evaluate the displacement of the object on the time interval $[0, 8]$. Sketch the graph of the velocity and shade the relevant region.
[3]
- (d) Compute the displacement on the interval $[0, 8]$ using antiderivatives of the function v .

Problem 4

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Problem 4

Problem 4 The **velocity** function for a man walking along a straight road which runs east and west is given by $v(t) = -t^2 + 4t - 3$ ft/min.

- (a) Set up a definite integral for the man's **displacement** during the time interval from 2 minutes to 6 minutes after he began running.
- (b) **At home:** Evaluate the definite integral using the limit of a right Riemann sum.
- (c) Is this the same as the total **distance** the man walked from 2 minutes to 6 minutes? Why or why not?